

Equation Source Audit

AI-Powered Unified Mobility and Delivery System Using Dynamic Feasibility Analysis

Supporting note — not part of the submitted manuscript

1. Question addressed

Which published paper is each of the twenty equations taken from, and on which page?

2. Method

Every equation-defining passage in the 13-page master manuscript was located by its mathematical anchor ($CS(i, j)$, $T(i, j)$, $V(i, j)$, $Q(i, j)$, the routing objective, $Cost(d, p)$, d^* , the boundedness and Lipschitz statements, Proposition 1, the complexity bound, the linear assignment problem, the softmax, the loss, the gradient terms, the update rule, the threshold utility, and the CCI definition). The 300 characters following each anchor were then searched for bracketed citation markers.

3. Finding

No equation in the master manuscript carries a citation. The scan returned zero reference markers against all twenty anchors. The only bracketed matches recovered were occurrences of the interval $[0, 1]$, which is notation rather than a reference.

Consequently there is no “taken from paper X, page Y” record to report for these equations, because no such attribution was ever made. The relations were not lifted from particular articles; they are standard constructions instantiated for this problem, together with several relations defined and proved in the paper itself.

4. The one attribution that can be made with a page range

Eq.	Source	Location
(14)	H. W. Kuhn, “The Hungarian method for the assignment problem,” <i>Naval Research Logistics Quarterly</i> , vol. 2, no. 1–2, 1955, doi: 10.1002/nav.3800020109	pp. 83–97 (whole article); reference [12] in the manuscript

This is the sole equation in the paper with a named originator. It is already cited in Section III-D and appears in the reference list.

5. Per-equation status

Eq.	Statement	Source cited in master	Actual status
(1)	Capability-coverage index	None	Defined in this paper as an evaluation device for Table I; not claimed as a contribution
(2)	Compatibility score $CS(i, j)$	None	Weighted linear scalarisation (standard in multi-criteria decision analysis); the five criteria and their use for tri-modal dispatch are this paper’s
(3)	Admission rule $CS \geq \tau$	None	Defined in this paper
(4)	Time-window overlap $T(i, j)$	None	Triangular similarity kernel (standard); the per-type tolerance W_{type} is contributed here
(5)	Capacity indicator $V(i, j)$	None	Capacity feasibility indicator conventional in capacitated vehicle routing
(6)	Priority compatibility $Q(i, j)$	None	Normalised-difference similarity measure (standard)
(7)	Route objective	None	Weighted-sum scalarisation, conventional for pickup-and-delivery with time windows
(8)	Assignment cost	None	Weighted linear cost (standard)
(9)	Greedy driver selection	None	$\arg \min$ over available drivers (trivial)
(10)	Boundedness $CS \in [0, 1]$	None	Proved in this paper from convexity of (2)
(11)	Lipschitz bound on T	None	Lipschitz continuity (classical analysis) applied to (4); the service-type asymmetry is this paper’s result
(12)	Fail-closed bound (Prop. 1)	None	Proved in this paper
(13)	Complexity $O(\mathcal{R} k)$	None	Bounded-neighbourhood argument stated in this paper
(14)	Linear assignment problem	None in master; now cited as [12]	Kuhn, 1955, pp. 83–97
(15)	Softmax parameterisation	None	Standard machine-learning construction
(16)	Binary cross-entropy loss	None	Standard classification loss
(17)	$\partial CS / \partial \alpha_k = F_k$	None	Derivative of the linear form (2) (trivial)
(18)	Softmax Jacobian	None	Standard textbook result
(19)	Gradient update on θ	None	Stochastic gradient descent (standard)
(20)	Threshold utility $\mathbb{E}[U(\tau)]$	None	Expected utility of a decision threshold, as in ROC analysis (standard)

6. Why page numbers cannot be supplied for the remainder

Softmax, cross-entropy, gradient descent, weighted-sum scalarisation, capacity indicators and Lipschitz continuity are common apparatus in their respective fields. They appear across a large body of literature and in standard textbooks, and no single article is their source. Attaching a specific paper and page to any of them would misrepresent where the formulation came from, and would fail immediately if the cited page were checked.

7. Options

(a) **Leave as is.** Section III-F of the manuscript already states which relations are original, which are standard, and which is classical and cited. This is accurate and defensible.

(b) **Add canonical texts for the standard constructions.** Two or three textbook references may be added and cited in the form “for the standard treatment, see [X]” — not “taken from [X]”. Suitable coverage would be a machine-learning text for (15), (16), (18) and (19); a multiobjective optimisation text for (7) and (8); and a vehicle-routing text for (5). This would raise the reference count from 12 to roughly 15.

8. Position to take in review or viva

No new mathematics is claimed. What is claimed is the feasibility-first ordering of the decision, the five-criterion score spanning three service types with per-type timing tolerance, and the three properties proved in Section III-D — boundedness, the Lipschitz asymmetry, and the fail-closed bound of Proposition 1. Stated in those terms the contribution is accurate and withstands checking; a claim over the individual equations would not.